LEDGER · 2026-09-04 · ELECTROCHEMISTRY / BIOSENSING

Laser-Induced Porous Graphene (LIG) Sensor Substrates via Direct-Laser Writing

REJECTED

This concept did **not** clear the framework.

The reasoning is published in full below.

What the assessment found

Incumbent benchmark	Metrohm DropSens DRP-110 screen-printed carbon electrode
Entry application	Point-of-Care Food Safety and Veterinary Residue Detection
Measured advantage	2.87-fold higher analytical sensitivity for sulfadimidine detection and 475% increase in peak current for nitrite oxidation compared to Metrohm DropSens DRP-110.
Time to first revenue	9 months
Gross margin at parity	95.5%
Startup CapEx	$45,000
Payback from first sale	0.1 months
Capital productivity	192.0x

Regulatory risk	Moderate — qualification costs reported ($75,000 estimated)

Why it failed

- Voided by: Mismatched comparison, Form-factor / handling regression, Absence search missing
- Evidence rests on non-peer-reviewed sources
- Absence asserted, search unreported

Assessment

The concept exhibits strong margins and clear scientific performance advantage, but fails due to a severe form-factor regression (flexible polyimide vs rigid ceramic card) requiring unproven interface adapters. Additionally, it lacks a structured Market Absence Ledger section with quantitative company/product search counts.

Also flagged

- Price parity asserted, not sourced
- runsheet_no_quantities

A rejection is not a claim that the science is wrong. It means the venture case did not survive: the comparator was not what the buyer actually buys, the comparison was not like-for-like, the advantage fell short of a step change, or the capital and time to first revenue were prohibitive.

Assessed 2026-09-04 against seven gates plus the voiding sub-tests. How the method works · Browse all concepts